

Education

- 2021 – present ♦ **Ph.D. Computer Science**, Max Planck Institute for Intelligent Systems & University of Tübingen, Advisor: Prof. Georg Martius. *[Expected graduation: 2026]*
- Intrinsically Motivated Open-ended Learning, Unsupervised Exploration in Reinforcement Learning (RL), World Models and Self-improvement at Scale.
- 2018 – 2021 ♦ **M.Sc. Electrical Engineering and Information Technology**, Technical University of Munich.
- Graduated with High Distinction, GPA 4.0/4.0 (German grading system: 1.0)
- Specialization in Robotics and Machine Learning
- 2015 – 2018 ♦ **B.Sc. Electrical Engineering and Information Technology**, Technical University of Munich.
- Graduated with High Distinction, GPA 4.0/4.0 (German grading system: 1.0)
- 2010 – 2015 ♦ **German Foreign High School, Abitur Diploma**, Istanbul Lisesi.
- GPA 4.0/4.0 (German grading system: 1.0), Graduated as top of my class.

Work Experience

- April – Sep 2025 ♦ **Research Internship**, Meta FAIR, Paris — CodeGen Team.
- Scaling RL for LLM code generation: synthetic data generation, curriculum design, and self-play across multiple model families.
- July – Nov 2024 ♦ **Research Internship**, Qualcomm AI Research, Amsterdam.
- Investigated hybrid reasoning strategies to improve robustness and generalization in Vision-Language-Action models.
- Apr 2017 &
Sep–Oct 2017 ♦ **Research Engineering Internship**, Intel, Munich.
- Computational cost estimation of machine learning algorithms for LTE modem power optimization

Publications

Cansu Sancaktar, David Zhang, Gabriel Synnaeve and Taco Cohen. **A Deep Dive into Scaling RL for Code Generation with Synthetic Data and Curricula**, **Preprint, 2026**.

Swadesh Jana*, **Cansu Sancaktar***, Tomáš Daniš, Georg Martius, Antonio Orvieto and Pavel Kolev. **GASP: Guided Asymmetric Self-Play For Coding LLMs**, **ICLR 2026 Workshop on AI with Recursive Self-Improvement (Spotlight)**

Pietro Mazzaglia, **Cansu Sancaktar**, Markus Peschl and Daniel Dijkman. **Hybrid Training for Vision-Language-Action Models**, **ICLR 2026**.

Cansu Sancaktar*, Christian Gumbsch*, Andrii Zadaianchuk, Pavel Kolev and Georg Martius. **SENSEI: Semantic Exploration Guided by Foundation Models to Learn Versatile World Models**, **ICML 2025**. [Project Page]

Jiaqi Chen, Ji Shi, **Cansu Sancaktar**, Jonas Frey and Georg Martius. **Offline vs. Online Learning in Model-based RL: Lessons for Data Collection Strategies**, **RLC 2025**.

Albane Ruaud, **Cansu Sancaktar**, Marco Bagatella, Christoph Ratzke and Georg Martius. **Modelling Microbial Communities with Graph Neural Networks**, **ICML 2024**. [Project Page]

Cansu Sancaktar, Justus Piater and Georg Martius. **Regularity as Intrinsic Reward for Free Play**, **NeurIPS 2023**. [Project Page, Code]

Bhavya Sukhija, Lenart Treven, **Cansu Sancaktar**, Sebastian Blaes, Stelian Coros and Andreas Krause. **Optimistic Active Exploration of Dynamical Systems**, **NeurIPS 2023**.

Cansu Sancaktar, Sebastian Blaes and Georg Martius. *Curious Exploration via Structured World Models Yields Zero-Shot Object Manipulation*, **NeurIPS 2022**. [Code]

- Best poster award at the IEEE RAS Model-Based Optimization for Robotics poster event, 2022.

Nico Gürtler, Felix Widmaier, **Cansu Sancaktar**, ... and Georg Martius, *Real robot challenge 2022: Learning dexterous manipulation from offline data in the real world.*, **NeurIPS 2022 Competition Track**.

Cansu Sancaktar, Marcel van Gerven, and Pablo Lanillos. *End-to-End Pixel-Based Deep Active Inference for Body Perception and Action*, 10th International Conference on Development and Learning and Epigenetic Robotics (ICDL-EpiRob), IEEE, 2020. [Poster presentation, Code]

Honors & Awards

- 2024 ♦ **MPI-IS Outstanding Female Doctoral Student Prize: Honorable Mention.**
- 2017 – 2021 ♦ **Scholarship Holder of the Max Weber-Program.**
This program aims at highly gifted students at universities in Bavaria.
- 2015 – 2020 ♦ **DAAD Scholarship Holder.**
A merit-based scholarship granted for my studies in Germany.
- Aug 2014 ♦ **Selected Attendee at the 8th Asian Science Camp.**
Chosen as one of the 8 representatives of Turkey at the 8th Asian Science Camp which is an event organized by Nobel laureates.

Professional Activities

Workshops & Competitions

- 2023 & 2024 ♦ Co-organizer of the workshop **Intrinsically Motivated Open-ended Learning**, **NeurIPS 2023 & 2024**.
- 2022 ♦ Co-organizer of the competition **Real Robot Challenge III - Learning Dexterous Manipulation from Offline Data in the Real World**, **NeurIPS 2022**.

Outreach

- 2022 – 2024 ♦ Co-organizer of the S4 Seminar Series of the IMPRS-IS graduate program.
- 2022 - 2023 ♦ Elected student representative of the IMPRS-IS graduate program for MPI-IS Tübingen.
- 2021 - 2022 ♦ Co-organizer of the Talk & Talk series at the Max Planck Institute for Intelligent Systems.

Teaching & Supervision

- 2024 ♦ **Jiaqi Chen, ETH Zürich**, Master's Thesis: *Offline vs. Online Learning in Model-based RL: Lessons for Data Collection Strategies* (accepted at RLC 2025)
- ♦ **Pulkit Goyal, University of Tübingen**, Master's Thesis: *Building Visual Semantic Bias in Curious Exploration during Free Play*.
- 2023 ♦ **Pro-seminar in Reinforcement Learning**, University of Tübingen.
- ♦ **Pulkit Goyal, University of Tübingen**, Essay Rotation: *Can Self-Exploring (Curious) RL Agents Model OCD?*
- ♦ **Shukrullo Nazirjonov, CaCTüS Internship**, *Extending Intrinsically Motivated Reinforcement Learning to Real Robots*.

Theses

- 2021 ♦ **Master's Thesis: State-Space Models for Discovering Low-Dimensional Dynamics in Neurophysiological Recordings**
► Advisor: Prof. Jakob Macke, *Machine Learning in Science*, University of Tübingen.
- 2018 ♦ **Bachelor's Thesis: Long Short-Term Memory Networks as Adaptive Filters**
► Advisor: Prof. Wolfgang Utschick, *Methods of Signal Processing*, TUM.

Skills

- Languages ◇ Turkish (native), English (C2), German (C2), Korean (A1).
- Programming ◇ Python, C, C++, MATLAB, HTML, Linux/Shell Script.
- Frameworks ◇ PyTorch, JAX, Tensorflow, Keras.
- Robotics ◇ ROS, Gazebo, Arduino.
- Misc. ◇ $\LaTeX$, Inkscape, Simulink.

Invited Talks

- ◇ Phillip Isola's Lab @ MIT, *Nov 2025*.
- ◇ BeNeRL Seminar Series, *Nov 2023*.
- ◇ Human and Machine Cognition Lab (Charley Wu) at the University of Tübingen, *Nov 2023*.
- ◇ Computational Principles of Intelligence lab (Eric Schulz) at Max Planck Institute for Biological Cybernetics, *Mar 2023*.
- ◇ Scientific talk at the 2023 IMPRS-IS Interview Symposium, *Jan 2023*.